Linux Handheld Transceiver (LinHT)

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Disclaimer(s)

- I am not one of the developers…
  - So I’m not up to the minute on developments
- I’m (*eagerly*) following LinHT developments as they are reported by the developers
- LinHT is not a commercial, or grant-funded project; developers work on it *as-available*

Review of LinHT

Update for ZRDC 2026

- First practical, working, reproducible, *user alterable* open source *Software Defined Transceiver* for Amateur Radio
- Reasonable chance of wide availability
- Runs Linux OS and GNU Radio flowgraphs
- All info on m17project.org/linht-hw/

Review of LinHT

Update for ZRDC 2026

- *Very clever hack* by the developers to use a commonly available VHF / UHF FM portable transceiver as a “donor chassis”
- Retevis C62 (shown “stock”)



Review of LinHT

Update for ZRDC 2026

- Using a “donor chassis” means all the things that are hard for Amateur Radio development are already designed:
  - Enclosure, display, keyboard, battery, accessories are already designed and available
  - LinHT is a replacement circuit board for C62

Review of LinHT

Update for ZRDC 2026

- ***The Biggest Deal…***
- It’s working! Proven! Transmits! Receives! Different modes!
- Unit shown is LinHT display



Review of LinHT

Update for ZRDC 2026

- Modes *so far*:
  - FM transmission and reception with pre-/de-emphasis and CTCSS
  - SSB transmission and reception
  - M17 digital voice / data(?) transmission and reception
  - TETRA digital voice - reception only
  - Demo 64QAM transmission at 2 Mbps

Review of LinHT

Update for ZRDC 2026

- ***The Second Biggest Deal…***
- It runs GNU Radio flowgraphs; create a new mode or modulation in GNU Radio, and
  - *Run it in LinHT*
- ***Third biggest deal*** - it's running (Yocto) Linux
  - SSH into your LinHT

Review of LinHT

Update for ZRDC 2026

- ***Three Prime (heroic) Developers***
  - Andreas Schmidberger OE3ANC - Yocto, Linux integration, device tree, drivers, GUI integration, and much of the Rev B hardware testing
  - Vlastimil Slintak OK5VAS - PCB design, manufacturing preparation, assembly coordination, first bring-up, ATtiny PMU work, GNSS tests, and Rev C hardware fixes.
  - Wojciech Kaczmarski SP5WWP - M17 portions, RF / DSP direction, public updates, gr-m17/libm17, PA interpretation, technical review
- All European - Austria, Czech Republic, Poland

Review of LinHT

Update for ZRDC 2026

- ***Not done yet***
- Rev B has been proven out, with some issues:
  - The GNSS ceramic antenna too close to the aluminium chassis (and in some positions touched it)
  - USB issues
  - *Many* other items on the to-do list

LinHT Next Revision - Rev C

Update for ZRDC 2026

- GNSS system will be working
  - (**G**lobal **N**avigation **S**atellite **S**ystem) - GPS (US) Galileo (EU)
- USB should be solid
- Most “transplant” components now sourced
  - Manufactured Rev C shouldn’t have to transplant most components (except display?)

LinHT Next Revision - Rev C

Update for ZRDC 2026

- ***As I understand it…***
- Rev C should be able to be sent to PCB assembly for “mass” production
  - I’ll guess hundred(s), not thousands
- Distribution method, cost, source country TBD
- Buyer will disassemble a C62, transplant LinHT PCB

LinHT Next Revision - Rev C

Update for ZRDC 2026

- ***Commercial version of LinHT?***
- Not anytime soon
  - Problem 1 - China manufacturers are “copycat”, very averse to doing original engineering and development
    - Some don’t do *any* - just manufacture existing designs
  - Problem 2 - Some manufacturers need to see some actual units working that it’s a viable unit
  - Problem 3 - Open Source - “there be dragons” (for big manufacturers)

LinHT Next Revision - Rev C

Update for ZRDC 2026

- ***LinHT has been, and remains, a low budget Amateur Radio project - no grant funding***
- It's formally a project of the M17 Foundation, with some recurring donations
- Besides development expense, M17F is trying to promote LinHT such as exhibiting at HAM RADIO (Friedrichshafen)
- Please consider donating to help finish LinHT: m17project.org, **Donate** link in upper right corner

LinHT - Big Picture in Amateur Radio

Update for ZRDC 2026

- ***LinHT is the most important hardware project in Amateur Radio today*.***
  - Bruce Perens K6BP 2025-08-12
- ***I agree with K6BP!***

* web.archive.org/web/20251104201636/https://perens.com/2025/08/12/whats-wrong-with-ardc/

LinHT - Big Picture in Amateur Radio

Update for ZRDC 2026

- Amateur Radio is conditioned not to think that *Software Defined* Radio doesn't actually mean that. And usually SDR means SD *Receiver*.
- Minor example - a 70 cm "all mode" radio, when you want to use SSB is an "all mode" radio that you have to buy with that functionality - and $$$
  - Example - Icom IC-9700 - $1900 (100 watts)

LinHT - Big Picture in Amateur Radio

Update for ZRDC 2026

- We need Amateur Radio's general perception of SDR to change to…
- ***Any Mode Radio***
  - And you as the user get to pick the modes that your radio will operate - just download software.
- No different than buying a computer and choosing the OS and apps - spreadsheet, Internet, or music composition

LinHT - Big Picture in Amateur Radio

Update for ZRDC 2026

- We need a working example to “show and tell”
- LinHT isn’t limited to the portable form factor
  - Well, *this* particular board is…
- Nothing preventing a Linux *Mobile* Transceiver (LinMT)
- Or, other bands like 6m or 2m or 1.25m

LinHT - Big Picture in Amateur Radio

Update for ZRDC 2026

- We need Amateur Radio's general perception of SDR to change to…
- ***Any Mode Radio!***
- ***Really! Any… Mode…***
  - And *you* as the user get to pick the modes that *your* radio will operate - just download software
  - No different than buying a computer and choosing the OS and apps - spreadsheet, Internet, or music composition